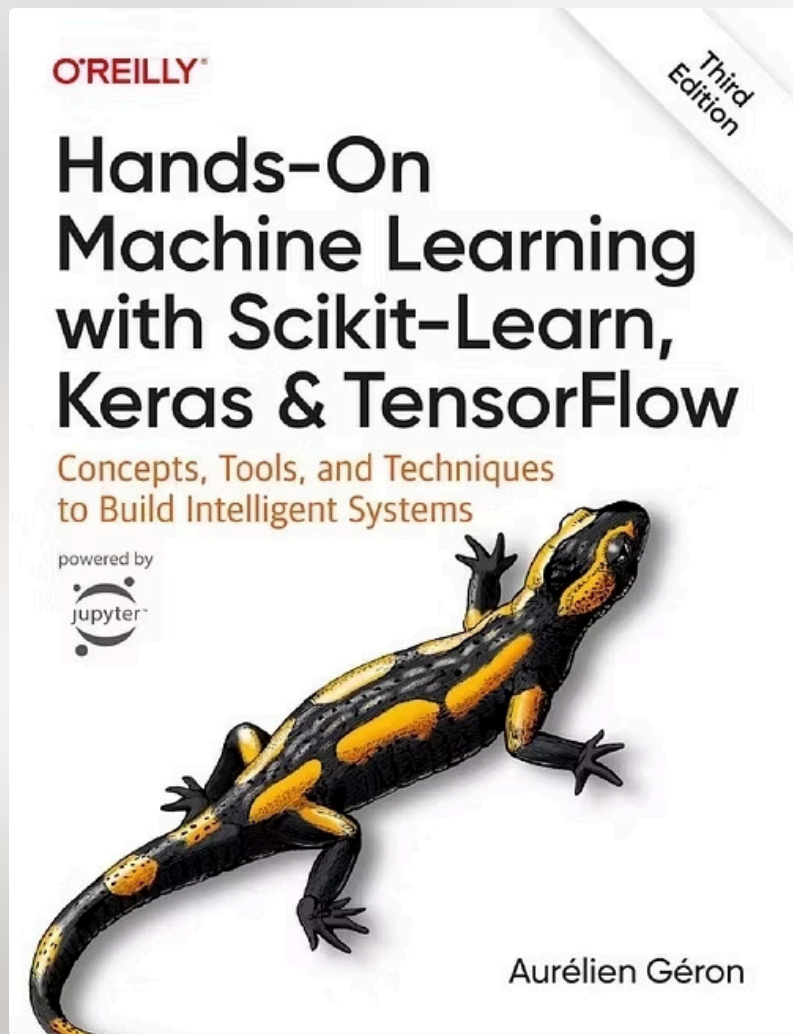




Machine Learning

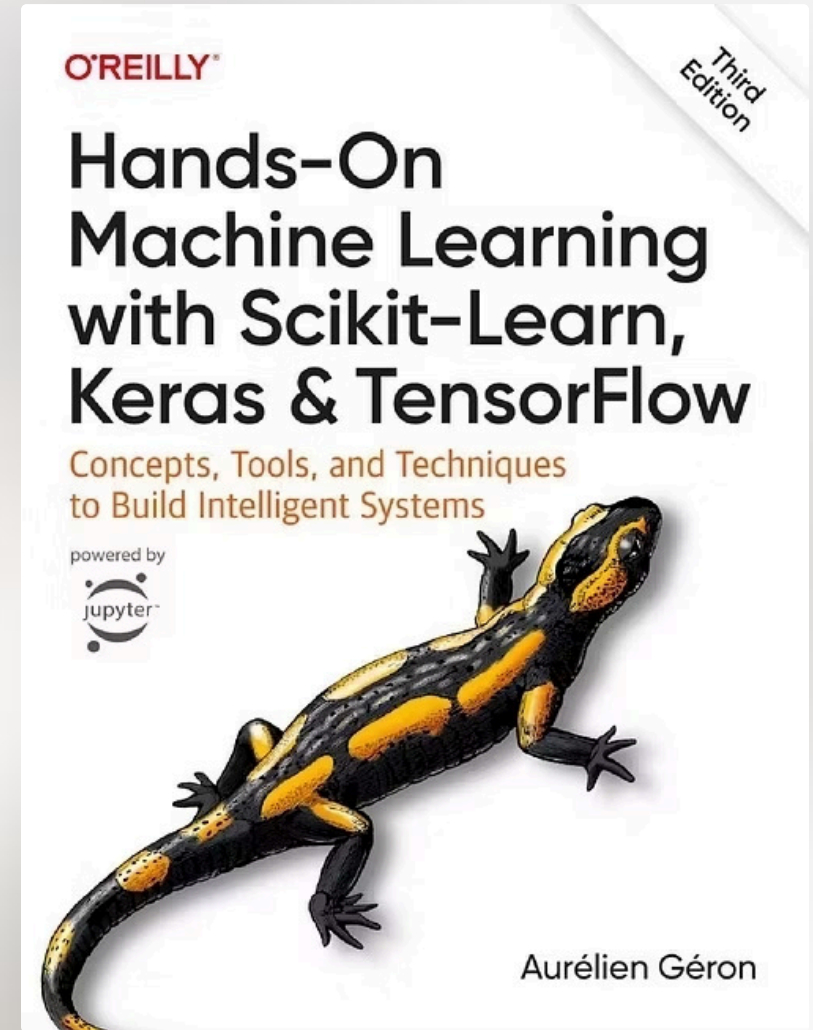
Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 08/07/2025

Machine Learning

Chapter 1 – The Machine Learning Landscape



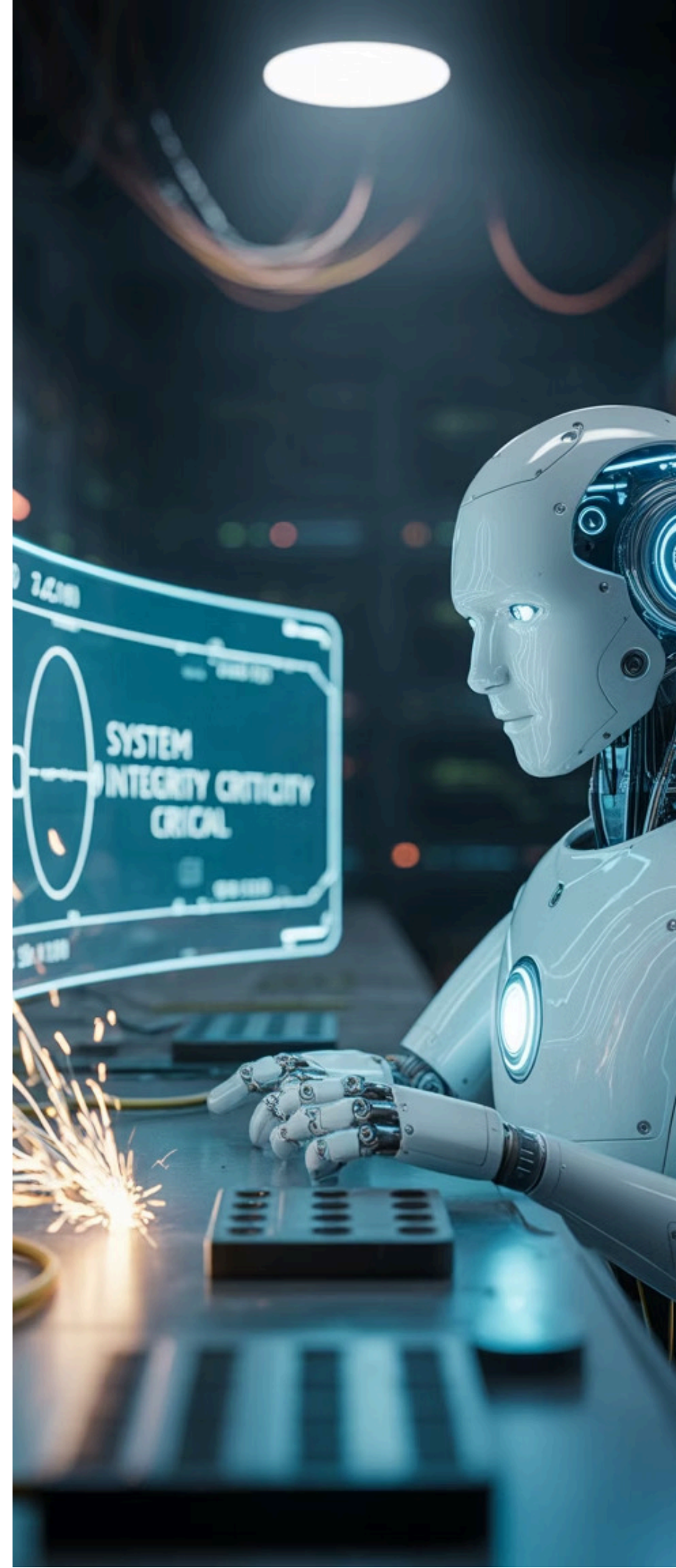


Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
5. **Main Challenges of Machine Learning**
6. Testing and Validating

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
 - **Classification criteria #1:** Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - **Classification criteria #2:** Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - **Classification criteria #3:** Learning Method
 - Instance-Based and Model-Based Learning
5. **Main Challenges of Machine Learning**
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - **Irrelevant Features**
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back: Reviewing the Chapter So Far
6. **Testing and Validating**
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch



Main Challenges of Machine Learning

Insufficient Quantity of Training Data, Non-representative Training Data, Poor-Quality Data, **Irrelevant Features**, Overfitting the Training Data, Underfitting the Training Data, Stepping Back: Reviewing the Chapter So Far



Irrelevant Features: Overview

Impact of Irrelevant Features

Garbage in, garbage out: If the training data contains irrelevant features, the system will **struggle to learn** effectively.

Critical Success Factor

The **quality** of the features used to train the model plays a **crucial** role in the **success** of a **ML project**.

Feature Engineering: The Key to Good Data

What is Feature Engineering?

Feature engineering is the **process** of **creating** and **selecting** the **right features** for training a **ML model**.

Steps in Feature Engineering:

Feature Selection

Choose the **most useful features** from the **available data**.

Feature Extraction

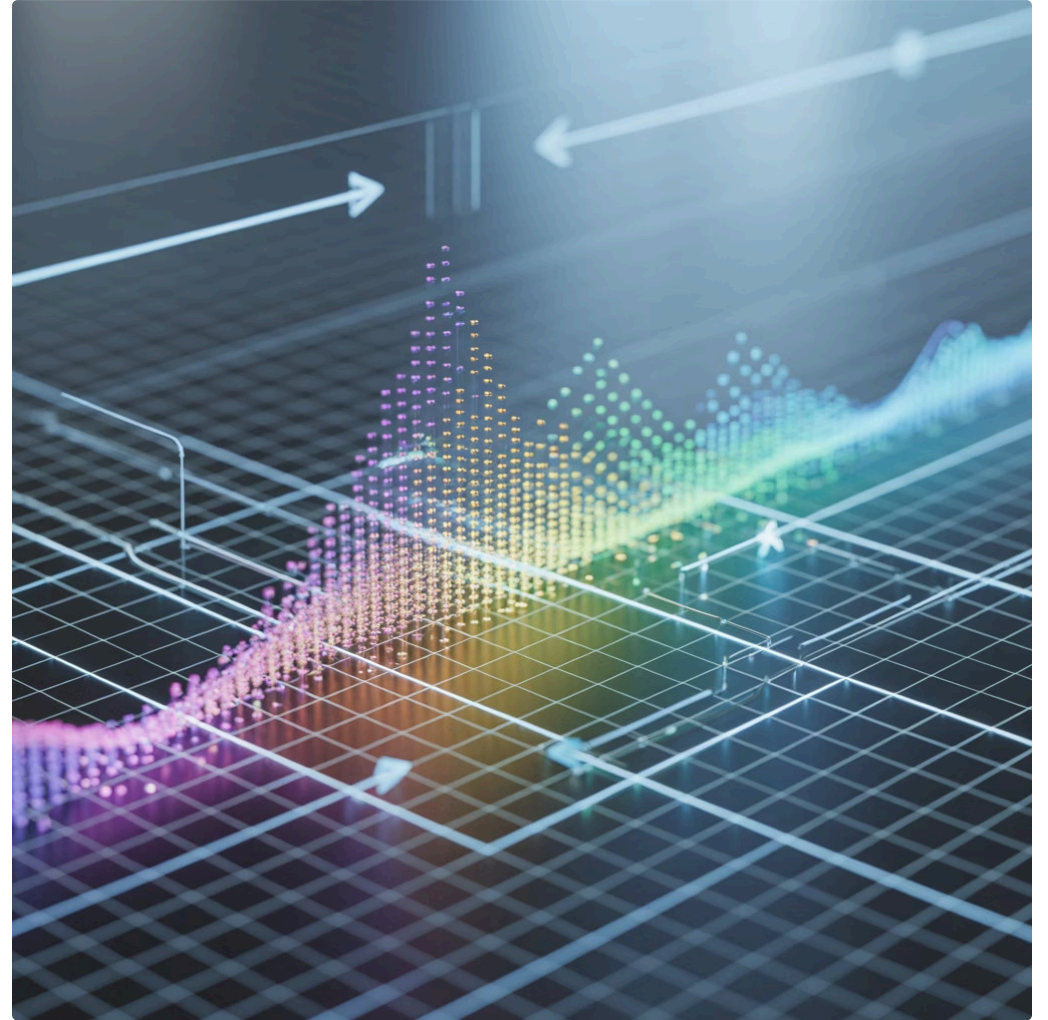
Combine or **transform** existing features into more useful ones (e.g., **dimensionality reduction algorithms**).

Creating New Features

Gather **additional data** to create **new features** that can provide more useful information.

Example: Dimensionality Reduction

- **Goal:** Reduce the number of **features** while **retaining** the most **important information**.
- **Example:** Principal Component Analysis (PCA) **combines features** to reduce complexity and retain relevant data for better model performance.



Key Takeaways



Importance of Relevant Features

The model's ability to **generalize** depends heavily on selecting **relevant features** and minimizing irrelevant ones.



Feature Engineering

Feature engineering involves **selection, extraction**, and the **creation of new features** to **enhance model performance**.



Avoiding Irrelevant Data

Too many **irrelevant features** can **hinder the learning process** and **reduce the model's performance**.



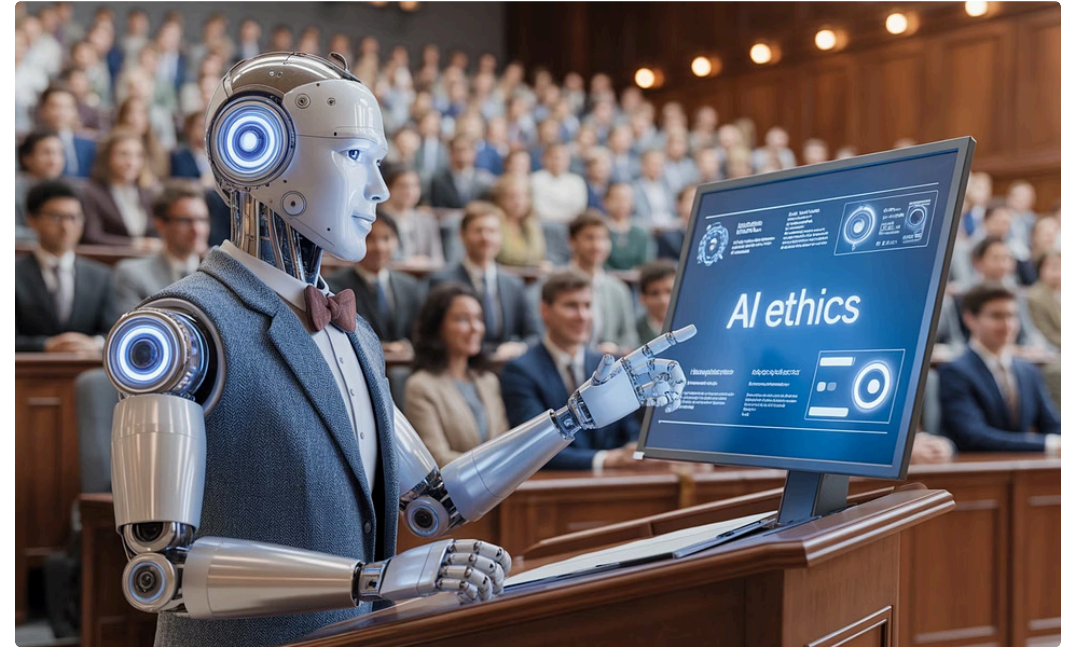
Next: Overfitting the Training Data

Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts